Electric car: launch the design

antocciphysics | TEACHER KEY | CPA

Learning target: Plan a safe team workflow and identify the car’s energy pathway.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 16; 3 course days later.*
No. Faraday’s law requires a change in flux, not merely nonzero flux.

02. Explain a worked example (worked)
3 m distance with clear timing boundaries; repeatable release and surface; speed distance/time.

03. Look and explain (model)
Accept a feasible labeled sketch with clear moving-part clearance.

04. Build the solution (practice)
Roles may rotate; teacher authorizes tool use and checks setup, secured work, appropriate eye protection and clear power-test area.

05. Use the idea (reason)
Example: complete the measured course within a chosen time while staying in lane; distance, time and lane observations provide evidence.

06. Choose a method (mixed) *Review: Unit 7, Day 7; 12 course days later.*
20 V; drops add to the supply voltage.

07. Exit check (exit)
Motor; teacher-approved wiring, secure parts, clear wheels and safe area.

08. STRETCH: Apply it another way (stretch)
Example finish within a marked lane on three runs; measure time and lateral deviation, then compare tradeoffs.

Teaching note: Build steps and safety approval follow the current classroom instructions. A written plan does not replace supervised tool training.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 16; 3 course days later.*
Grid effects depend on geometry, grounding, resistance, controls and spatially varying fields. A single loop does not represent the whole network.

02. Explain a worked example (worked)
3 m distance with clear timing boundaries; repeatable release and surface; speed distance/time.

03. Represent and interpret (model)
Accept a feasible labeled sketch with clear moving-part clearance.

04. Solve and justify (practice)
1.2 W input; motor heating, drivetrain friction, rolling losses, wheel rotation, sound and drag.

05. Analyze the reasoning (reason)
Example: complete the measured course within a chosen time while staying in lane; distance, time and lane observations provide evidence.

06. Choose a method (mixed) *Review: Unit 7, Day 7; 12 course days later.*
New resistance (4+2) R=6 R; current ratio 4/6≈0.6667.

07. Exit check (exit)
Measured course, consistent start/stop, repeated runs, matched battery/surface/setup.

08. STRETCH: Apply it another way (stretch)
Specific measurable targets and all-trial reporting; a prespecified mean/median and completion fraction avoid cherry-picking.

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